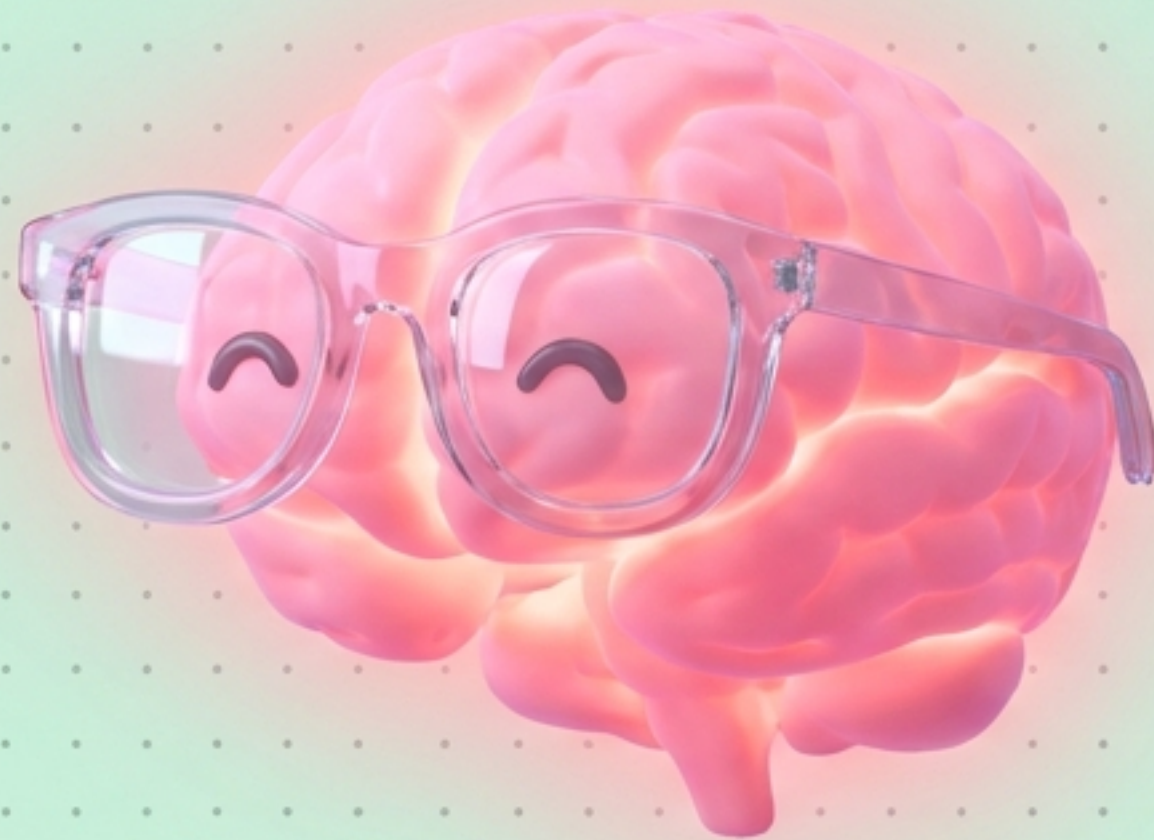


Stop scrolling!

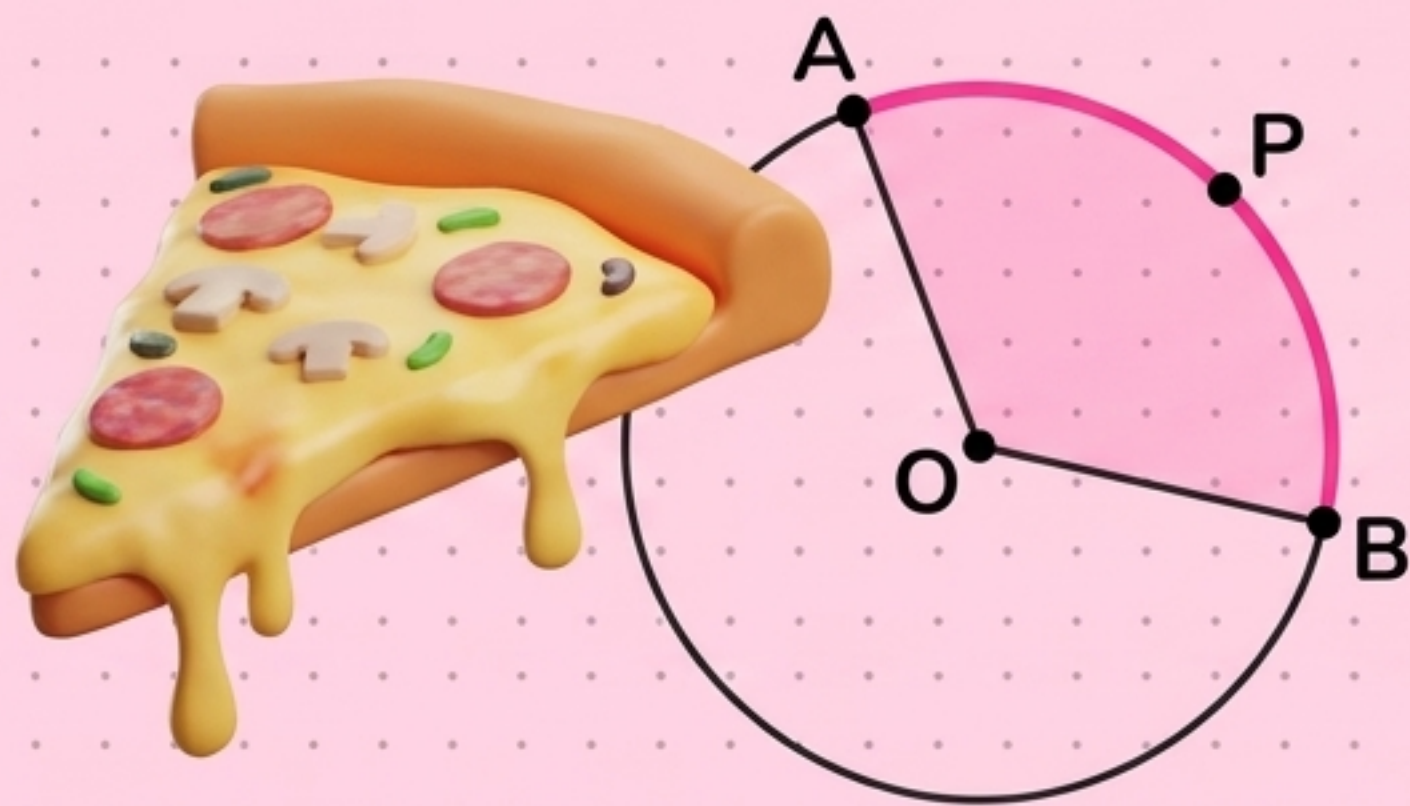


Let's master Areas Related to Circles in 5 minutes ✨

NCERT Class 10 Ch. 11 — Swipe to study 🖱️

The Sector

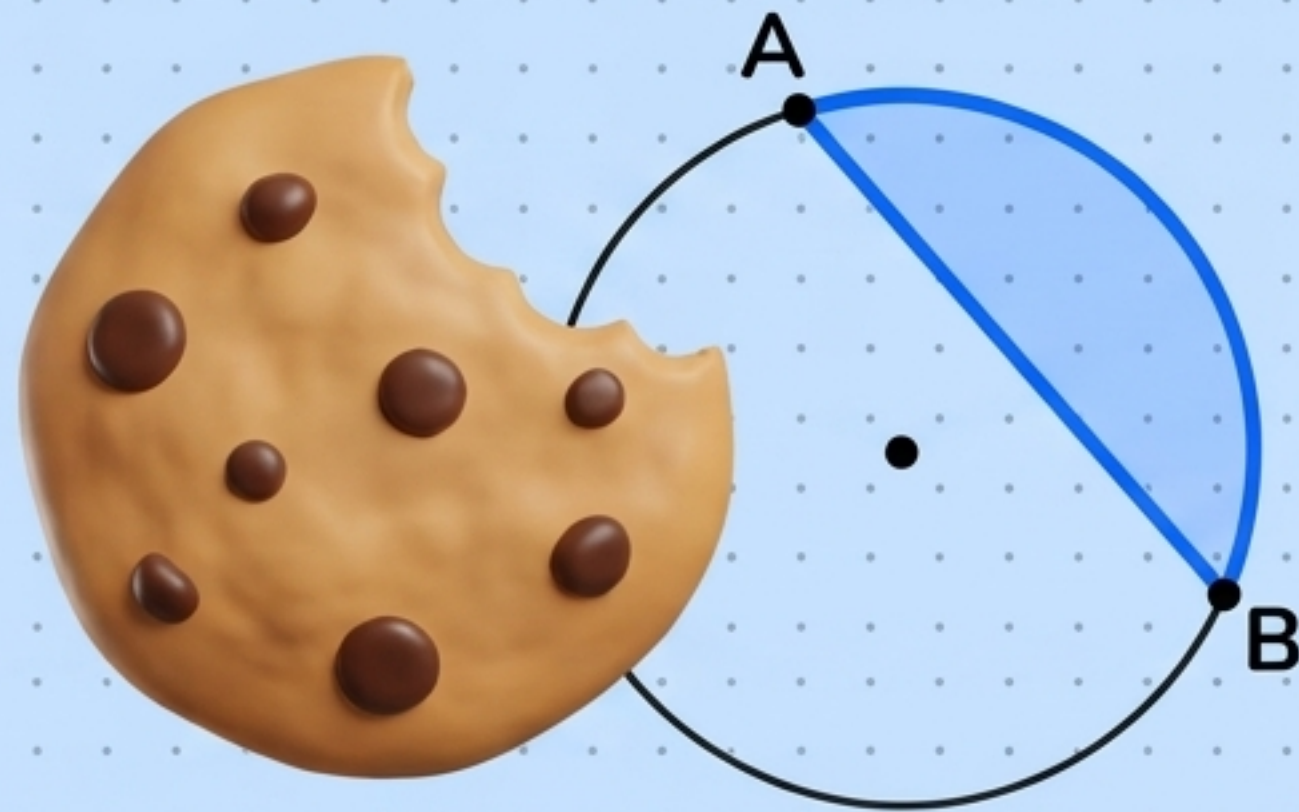
(Think: Pizza Slice 🍕)



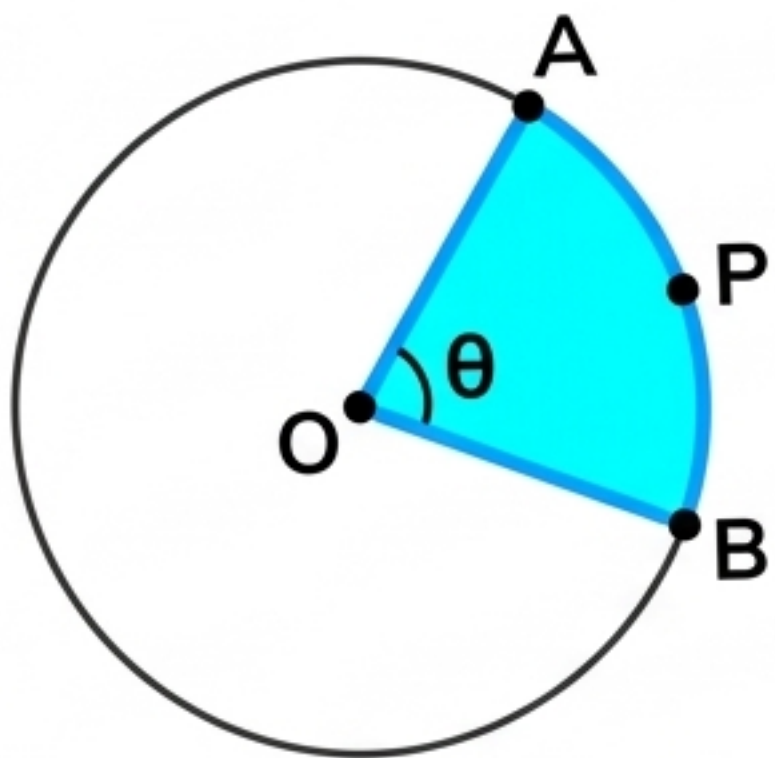
Enclosed by **2 Radii** +
the corresponding **Arc**.

The Segment

(Think: Cookie Edge 🍪)



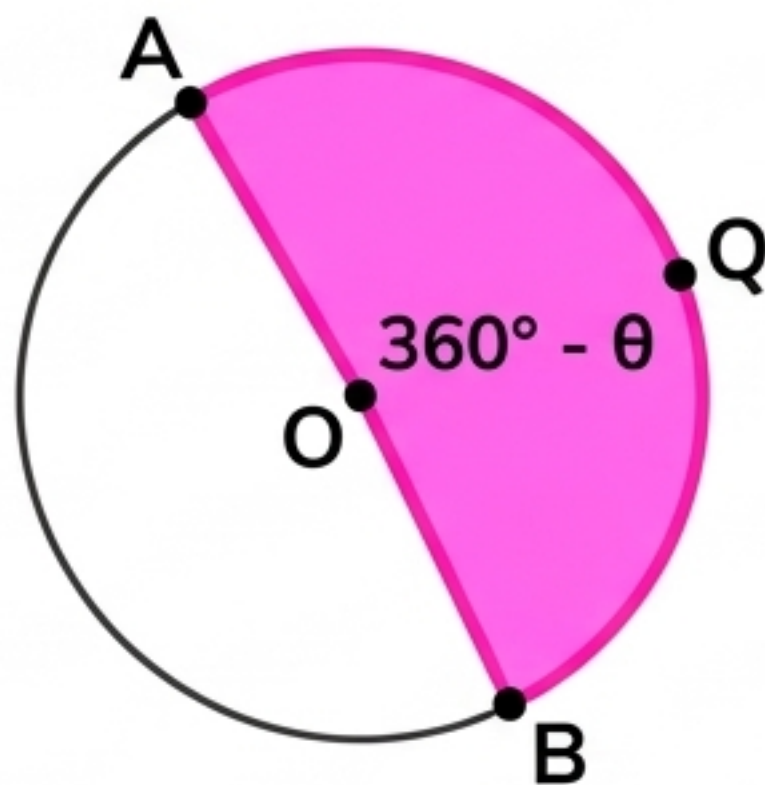
Enclosed by a **Chord** +
the corresponding **Arc**.



Minor Sector

The small slice! 🍰

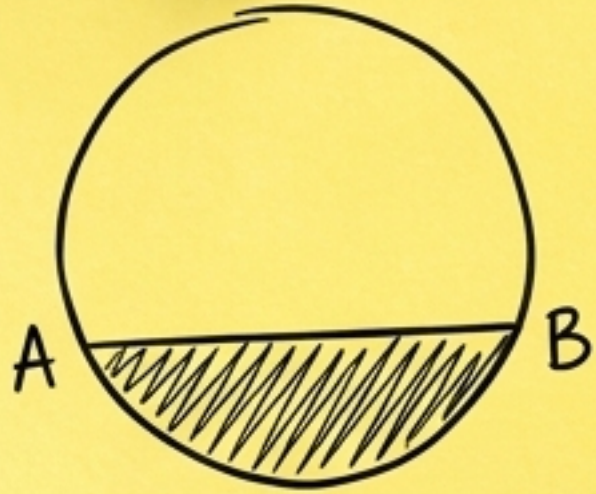
The angle at the center is θ .



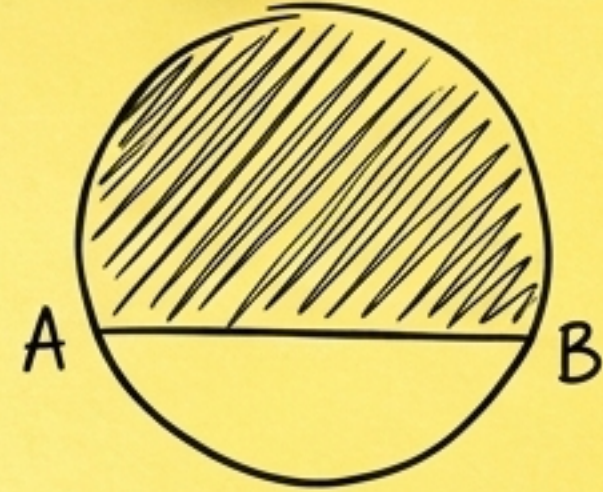
Major Sector

The rest of the pie! 🥧

The angle is the leftover: $360^\circ - \theta$.



Minor Segment: The small chunk chopped off by the chord AB.



Major Segment: The giant remaining chunk of the circle.

 **TEXTBOOK HACK:** If a question just says 'segment' or 'sector', they mean the MINOR one unless stated otherwise!

Whole
Circle = 360°

$$\text{Area} = \pi r^2$$

What if we just want 1 degree?

1° Slice

$$\text{Area} = \left(\frac{1}{360}\right) \times \pi r^2$$

Multiply by your
target angle (θ)!

θ° Slice

$$\text{Area} = \left(\frac{\theta}{360}\right) \times \pi r^2$$





Sector Area Formula

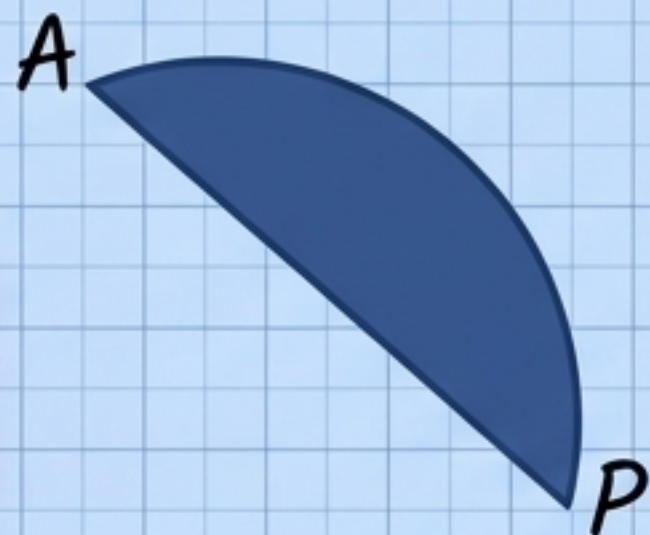
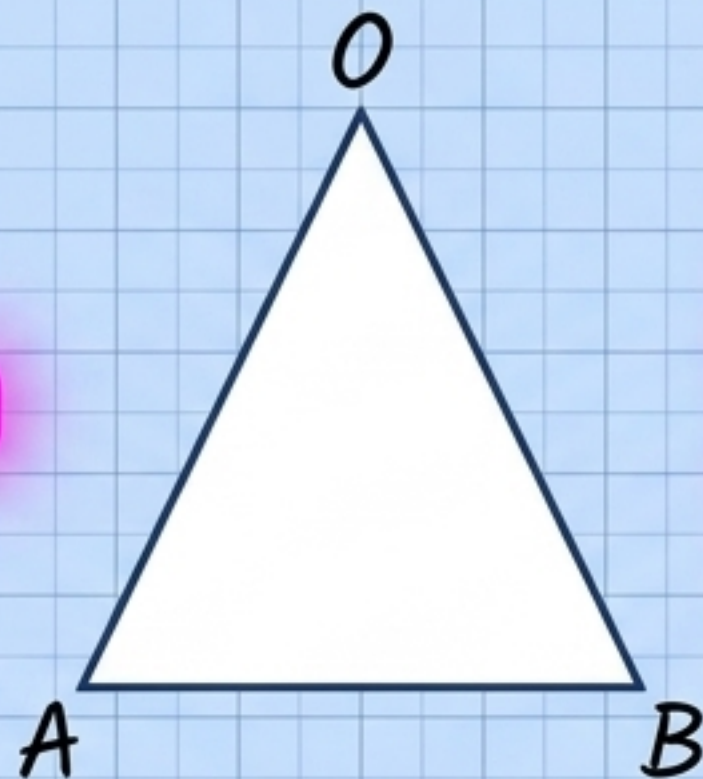
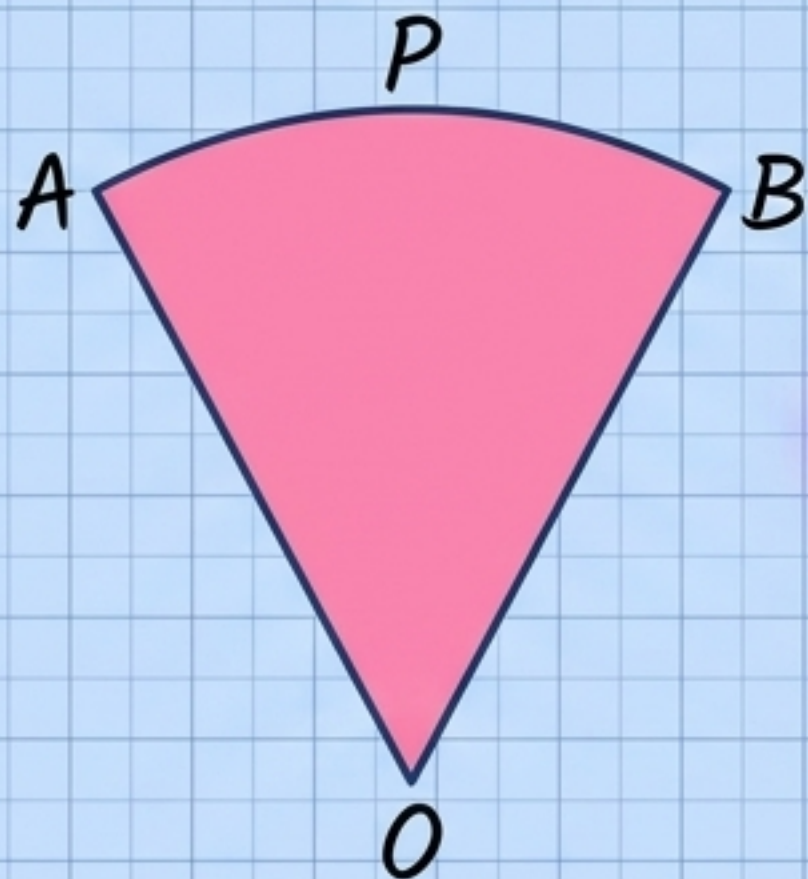
$$\text{Area} = \frac{\theta}{360} \times \pi r^2$$



Arc Length Formula

$$\text{Length} = \frac{\theta}{360} \times 2\pi r$$

The Segment Area Hack 💡

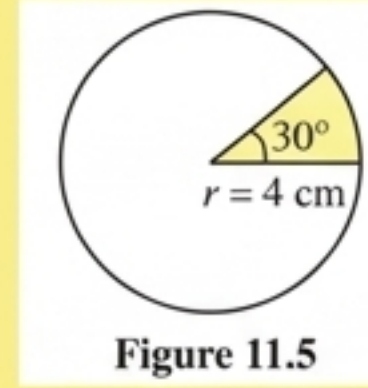


There's no direct formula!
Just find the area of the
pizza slice and subtract
the triangle. 🧠 ✨

Boss Fight! 🐙 (NCERT Ex 1)

Step 1

Given: Radius $r = 4$ cm,
Angle $\theta = 30^\circ$



Step 2

The Spell: Sector Area = $(\theta / 360) \times \pi r^2$

Step 3

$$\text{Area} = (30 / 360) \times 3.14 \times 4 \times 4$$

Step 4

Boom! 🎯 Final Answer = 4.19 cm^2



Arc Length

$$\left(\frac{\theta}{360}\right) \times 2\pi r$$

**Circle
Portions**

Sector Area

$$\left(\frac{\theta}{360}\right) \times \pi r^2$$

Segment Area

$$\text{Sector} - \text{Triangle}$$

Save this cheat sheet for your next math exam! 📌💖